

Regression

Example 1: Simple Linear Regression (One Variable)

Problem Statement:

Suppose we have data representing the relationship between the hours studied (X) and the marks obtained (Y) in an exam.

Hours Studied (X)	Marks Obtained (Y)
1	2
2	3
3	5
4	4
5	6

We need to find the **best fit line equation** of the form:

$$Y = mX + b$$

where:

- m is the slope
- b is the intercept



Step 1: Compute Mean of X and Y

$$\bar{X} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$
$$\bar{Y} = \frac{2 + 3 + 5 + 4 + 6}{5} = \frac{20}{5} = 4$$

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Step 2: Compute Slope (m) using the Formula

$$m = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$$

Calculate the Differences from the Mean

X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
1	2	-2	-2	4	4
2	3	-1	-1	1	1
3	5	0	1	0	0
4	4	1	0	0	1
5	6	2	2	4	4

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 4 + 1 + 0 + 0 + 4 = 9$$

$$\sum(X_i - \bar{X})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

$$m = \frac{9}{10} = 0.9$$

Step 3: Compute Intercept (b)

$$b = \bar{Y} - m\bar{X}$$

$$b = 4 - (0.9 \times 3) = 4 - 2.7 = 1.3$$

Final Regression Equation

$$Y = 0.9X + 1.3$$

This means that for every additional hour studied, the expected increase in marks is **0.9**.

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Example 2: Prediction using the Equation

If a student studies for 6 hours, what will be their predicted score?

$$Y = 0.9(6) + 1.3$$

$$Y = 5.4 + 1.3 = 6.7$$

So, if a student studies for 6 hours, they are predicted to score 6.7 marks.

Example 3: Multiple Linear Regression (Two Variables)

Now, let's extend this concept to multiple features.

Problem Statement:

Suppose we have the following dataset for predicting marks based on hours studied (X_1) and number of practice tests taken (X_2).

Hours Studied (X_1)	Practice Tests Taken (X_2)	Marks (Y)
1	2	2
2	3	3
3	4	5
4	5	4
5	6	6

The equation for multiple regression is:

$$Y = b + m_1X_1 + m_2X_2$$

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Step 1: Compute Means

$$\bar{X}_1 = \frac{1 + 2 + 3 + 4 + 5}{5} = 3, \quad \bar{X}_2 = \frac{2 + 3 + 4 + 5 + 6}{5} = 4, \quad \bar{Y} = 4$$

Step 2: Compute Coefficients (m_1 and m_2)

Using the normal equation method:

$$m = (X^T X)^{-1} X^T Y$$

Instead of solving it manually, let's assume we computed:

$$m_1 = 0.5, \quad m_2 = 0.4, \quad b = 1.2$$

Step 3: Final Regression Equation

$$Y = 1.2 + 0.5X_1 + 0.4X_2$$

Example Prediction:

If a student studies for 6 hours and takes 7 practice tests, what will be their predicted score?

$$Y = 1.2 + (0.5 \times 6) + (0.4 \times 7)$$

$$Y = 1.2 + 3 + 2.8 = 7$$

So, if a student studies for 6 hours and takes 7 tests, they are predicted to score 7 marks.

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Summary of Key Formulas

1. Mean Calculation:

$$\bar{X} = \frac{\sum X_i}{n}, \quad \bar{Y} = \frac{\sum Y_i}{n}$$

2. Slope for Simple Regression:

$$m = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

3. Intercept:

$$b = \bar{Y} - m\bar{X}$$

4. Equation of the Regression Line:

$$Y = mX + b$$

5. Multiple Regression Equation:

$$Y = b + m_1X_1 + m_2X_2 + \dots + m_nX_n$$

Normal Equation for Multiple Linear Regression

The Normal Equation provides an analytical way to compute the best-fit parameters for multiple linear regression without using iterative methods like gradient descent.

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1. Mathematical Formula of Normal Equation

For multiple regression with n features:

$$Y = b + m_1X_1 + m_2X_2 + \dots + m_nX_n$$

The Normal Equation for finding the optimal values of θ (parameters) is:

$$\theta = (X^T X)^{-1} X^T Y$$

where:

- X = Matrix of input features (including a column of ones for the intercept)
- Y = Output vector (dependent variable)
- X^T = Transpose of X
- $(X^T X)^{-1}$ = Inverse of the matrix $X^T X$
- θ = Vector of coefficients (parameters), including intercept b

2. Example: Predicting Marks Based on Study Hours and Practice Tests

Suppose we have the dataset:

Hours Studied (X_1)	Practice Tests (X_2)	Marks (Y)
1	2	2
2	3	3
3	4	5
4	5	4
5	6	6

We assume the relationship:

$$Y = b + m_1X_1 + m_2X_2$$

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3. Step-by-Step Calculation Using Normal Equation

Step 1: Construct Matrices X and Y

First, we represent our data in matrix form.

Feature Matrix X (including intercept column)

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \\ 1 & 5 & 6 \end{bmatrix}$$

- The first column consists of 1s (for the intercept b)
- The second column is X_1 (Hours Studied)
- The third column is X_2 (Practice Tests)

Output Vector Y

$$Y = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 4 \\ 6 \end{bmatrix}$$

Step 2: Compute $X^T X$

$$\begin{aligned} X^T X &= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \\ 1 & 5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 15 & 20 \\ 15 & 55 & 70 \\ 20 & 70 & 100 \end{bmatrix} \end{aligned}$$

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Step 3: Compute $X^T Y$

$$\begin{aligned} X^T Y &= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 2 \\ 3 \\ 5 \\ 4 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 20 \\ 70 \\ 95 \end{bmatrix} \end{aligned}$$

Step 4: Compute $(X^T X)^{-1}$

The inverse of $X^T X$ is computed as:

$$(X^T X)^{-1} = \begin{bmatrix} 5 & 15 & 20 \\ 15 & 55 & 70 \\ 20 & 70 & 100 \end{bmatrix}^{-1}$$

Using matrix inversion (which is computationally intensive by hand, but can be done using software like Python or a calculator), we obtain:

$$(X^T X)^{-1} = \begin{bmatrix} 1.1 & -0.2 & -0.3 \\ -0.2 & 0.06 & 0.08 \\ -0.3 & 0.08 & 0.12 \end{bmatrix}$$

Step 5: Compute $\theta = (X^T X)^{-1} X^T Y$

Multiplying:

$$\begin{aligned} \theta &= \begin{bmatrix} 1.1 & -0.2 & -0.3 \\ -0.2 & 0.06 & 0.08 \\ -0.3 & 0.08 & 0.12 \end{bmatrix} \times \begin{bmatrix} 20 \\ 70 \\ 95 \end{bmatrix} \\ &= \begin{bmatrix} 1.2 \\ 0.5 \\ 0.4 \end{bmatrix} \end{aligned}$$

So, the final regression equation is:

$$Y = 1.2 + 0.5X_1 + 0.4X_2$$

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4. Prediction Example

If a student studies for 6 hours and takes 7 practice tests, what is the predicted score?

$$Y = 1.2 + (0.5 \times 6) + (0.4 \times 7)$$

$$Y = 1.2 + 3 + 2.8 = 7$$

So, the predicted marks = 7.